

Let Token Draw Itself

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Abstract

Recent autoregressive image generation models achieve impressive results by modeling images as sequences of discrete tokens. However, these tokens are treated as opaque latent codes that are decoded jointly, making the visual contribution of each generation step invisible. We propose that each token should instead independently carry its own visual content as a self-contained drawing primitive. We present TokenPainter, where every token independently produces an RGB color and spatial mask via per-token heads, composited via softmax into the final image. Without auxiliary losses, softmax competition drives tokens to spontaneously specialize into complementary spatial regions, from broad silhouettes to fine details. To train TokenPainter, we introduce GroupPos, a GAN loss that normalizes discriminator scores within each group via z-score and applies a soft non-negative threshold. TokenPainter trained with a fixed number of tokens extrapolates to different token counts at inference while preserving subject identity, and distinct subsets of tokens recombine into structurally consistent outputs, demonstrating token-level modularity.

Introduction

A scene is understood not as a single undivided image but as a collection of parts, each with its own shape and location. A painter builds a portrait by layering these parts so that the silhouette comes first, the features follow, and the details arrive last, which makes the subject recognizable early and adds fidelity with each addition. This compositional nature of visual understanding suggests that an image generation process should likewise be decomposable into visible, meaningful units whose individual contributions can be inspected.

In both autoregressive generation and stroke-based rendering, the process by which an image comes into being remains hidden from view, either because the steps are opaque or because they depend on an external reference. In autoregressive (AR) image generation (Liao et al. 2026; Wang et al. 2025), tokens are discrete codes that are decoded jointly, which leaves the spatial role of each step invisible. A shared decoder mixes all tokens together, so the spatial footprint of any individual token cannot be recovered. Stroke-based rendering (Huang, Heng, and Zhou 2019; Ganin et al. 2018) makes the drawing process explicit but depends on reference images for supervision or an external renderer to translate actions into pixels.

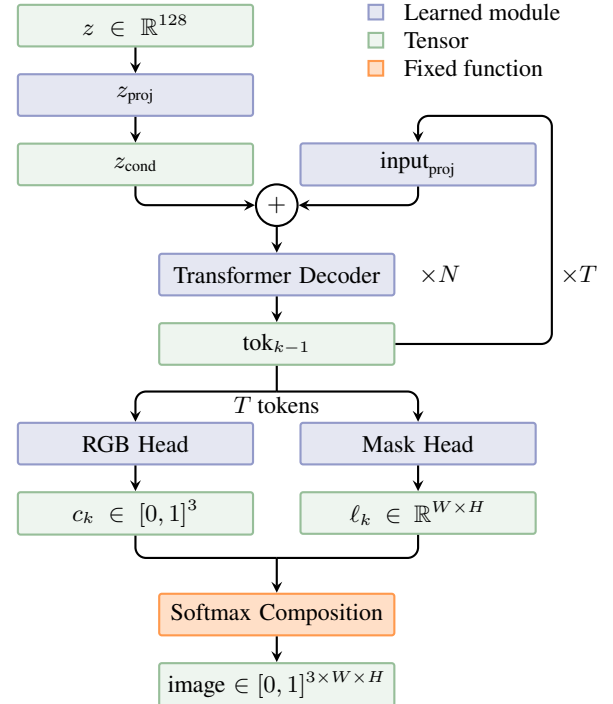


Figure 1: TokenPainter architecture. z is projected to z_{cond} and injected at every autoregressive step together with the preceding token. Each token is independently decoded into an RGB color and spatial logits, then composited via softmax into the final image.

We present **TokenPainter**, an autoregressive model where each token serves as a visible drawing primitive, so that every generation step is directly visible while requiring neither reference images nor an external renderer. Per-token heads independently decode each token into an RGB color and a spatial mask, and a softmax composite assembles the final image from these per-token elements. Because each token’s contribution is computed independently, every step in the generation sequence can be inspected in isolation, which makes the entire generation process interpretable. Without auxiliary losses or explicit decomposition objectives, softmax competition alone drives tokens to spontaneously spe-

cialize into complementary spatial roles, where some tokens cover broad silhouettes while others handle fine details. This emergent division of labor enables strong extensibility, where distinct subsets of tokens recombine into consistent outputs and the model extrapolates to different token counts at inference while preserving subject identity, revealing a surprising form of token-level modularity.

We train TokenPainter using a generative adversarial framework (Goodfellow et al. 2014), where a co-trained discriminator provides the sole learning signal. Inspired by group-based advantage normalization in GRPO (Yuan et al. 2025), we introduce GroupPos, a GAN loss that normalizes discriminator scores via a global z-score across the entire batch, and gates advantages through a soft non-negative function so that below-mean samples contribute small positive weight.

Our contributions are as follows.

- We present TokenPainter, in which per-token heads independently decode RGB color and spatial logits, turning each token into a visible drawing primitive and making every generation step interpretable.
- We show that per-token independent decoding combined with softmax competition drives tokens to specialize into complementary spatial regions without auxiliary losses. This design enables strong extensibility, where distinct subsets of tokens recombine into consistent outputs and the model extrapolates to different token counts while preserving subject identity.
- We introduce GroupPos, a GAN loss with global z-score advantage normalization and soft non-negative gating.

Related Work

Autoregressive image generation has advanced rapidly since VQGAN (Esser, Rombach, and Ommer 2021) first combined CNN tokenization with Transformer modeling. MaskGIT (Chang et al. 2022) introduced bidirectional masked token prediction for faster decoding. VAR (Tian et al. 2024) reformulated autoregression as next-scale prediction, surpassing diffusion models for the first time. LlamaGen (Sun et al. 2024) showed that vanilla language-model architectures applied to image tokens can match state-of-the-art quality. Despite these advances, tokens in all these methods are latent codes decoded jointly through a shared decoder, leaving the spatial contribution of each step invisible. This opacity persists in recent methods that fine-tune pretrained AR generators via reinforcement learning. VA- π (Liao et al. 2026) combines GRPO with variational inference, reducing FID from 14.36 to 7.65 in 25 minutes on 8 A100s. AR-GRPO (Yuan et al. 2025) applies GRPO to AR image generation with three reward signals. Selftok-Zero (Wang et al. 2025) proves that 2D spatial tokens violate the Bellman equation and uses AR-only tokens for effective RL. Adv-GRPO (Mao et al. 2026) adds an adversarial reward to GRPO with DINO foundation features serving as visual priors. In all of these, tokens remain abstract codes decoded jointly by a pretrained decoder.

Object-centric methods share our goal of decomposing images into independent visual components. SlotVAE (Wang

et al. 2023) uses slot attention for object-centric scene generation, and SlotDiffusion (Wu et al. 2023) extends this principle to diffusion-based models. These methods require iterative attention mechanisms and explicit reconstruction or diffusion objectives to learn the decomposition. In TokenPainter, per-token spatial specialization emerges from softmax competition alone, without auxiliary losses or explicit decomposition objectives.

Stroke-based rendering makes the generation process visible by building images from sequences of strokes. Learning to Paint (Huang, Heng, and Zhou 2019) uses DDPG with a differentiable neural renderer, and Compositional Neural Painter (Hu et al. 2023) separates painting location from painting content. Content Masked Loss (Schaldenbrand and Oh 2021) improves stroke planning via object-detection-guided reward shaping. These methods depend on reference images for supervision. The notable exception is SPIRAL (Ganin et al. 2018), which uses a GAN discriminator as the sole reward signal for reference-free RL painting, later improved by SPIRAL++ (Mellor et al. 2019) with temporal credit assignment and spectral normalization. TokenPainter combines per-step visibility, reference-free training, and emergent specialization within a single autoregressive framework, where per-token heads make each step directly visible, softmax competition drives spatial specialization, and the GAN objective eliminates the need for reference images.

Method

Architecture

Figure 1 gives an overview of TokenPainter. Given a latent noise vector $z \sim \mathcal{N}(0, I)^{128}$, TokenPainter autoregressively generates T tokens ($\text{tok}_0, \dots, \text{tok}_{T-1}$) through a causal Transformer decoder (Vaswani et al. 2017), where each token is a continuous latent vector rather than a discrete codebook index. Each token is independently decoded into an RGB color $c_k \in [0, 1]^3$ and spatial logits $\ell_k \in \mathbb{R}^{H \times W}$ via per-token heads. The decoded colors and masks are composited via a softmax-weighted convex combination to produce the final $H \times W$ image.

Latent encoding. We project the noise z through a 2-layer MLP to obtain a conditioning vector z_{cond} , which encodes the global generation intent and is summed into the Transformer input at every autoregressive step.

Autoregressive generation. Generation proceeds through T steps. At each step k , the Transformer input is formed by summing z_{cond} and the previous token’s output projected back into the input space through a 2-layer MLP $\text{input}_{\text{proj}}(\text{tok}_{k-1})$:

$$x_k = z_{\text{cond}} + \begin{cases} 0 & k = 0 \\ \text{input}_{\text{proj}}(\text{tok}_{k-1}) & k > 0 \end{cases} \quad (1)$$

At the first step ($k = 0$), only z_{cond} is fed to the Transformer, acting analogously to a start-of-sequence token that carries the global generation intent, while subsequent steps incorporate the feedback from the preceding token. The Transformer decoder consists of 4 blocks with causal multi-head self-attention and feed-forward layers with GELU activation (Hendrycks and Gimpel 2016). Rotary position encoding

(RoPE) (Su et al. 2024) is applied to the query and key within self-attention, and causal attention across steps is maintained via a KV-cache that accumulates key-value pairs from previous steps.

Token decoding. Each token representation tok_k is independently decoded through two per-token heads. The RGB Head is a 3-layer residual MLP with LayerNorm that produces $c_k \in [0, 1]^3$. The Mask Head expands the token through a fully-connected layer into a feature map, upsamples via PixelShuffle (Shi et al. 2016), and applies several refinement convolutions, producing logits $\ell_k \in \mathbb{R}^{H \times W}$. Both heads operate independently without cross-token interaction, forcing each token to be a self-contained visual primitive.

Softmax composition. The per-token colors c_k and logits ℓ_k are composited via softmax-normalized masks:

$$[m_1, \dots, m_T] = \text{softmax}([\ell_1, \dots, \ell_T]) \quad (2)$$

$$\text{image} = \sum_{k=1}^T m_k \cdot c_k \quad (3)$$

The convex combination ($\sum m_k = 1$ at every pixel) produces each pixel’s color as a weighted mixture of the token colors, where each token competes globally for every pixel via its mask logits.

GAN Training

We train TokenPainter using a generative adversarial framework (Goodfellow et al. 2014), where a co-trained discriminator provides the sole learning signal.

Discriminator loss. The discriminator is trained with a softplus loss and R1 and R2 gradient penalties (Mescheder, Geiger, and Nowozin 2018):

$$\mathcal{L}_D = \mathbb{E}_{\text{real}}[\text{softplus}(-D(x))] + \mathbb{E}_{\text{fake}}[\text{softplus}(D(\hat{x}))] + \gamma \cdot (R1 + R2), \quad (4)$$

where R1 penalizes the gradient of D with respect to real images, and R2 penalizes the gradient with respect to fake images. The discriminator uses a separate batch of random z with independent augmentation via an adaptive discriminator augmentation (ADA) pipeline (Karras et al. 2020), which adjusts augmentation strength based on the discriminator’s sign accuracy to prevent overfitting.

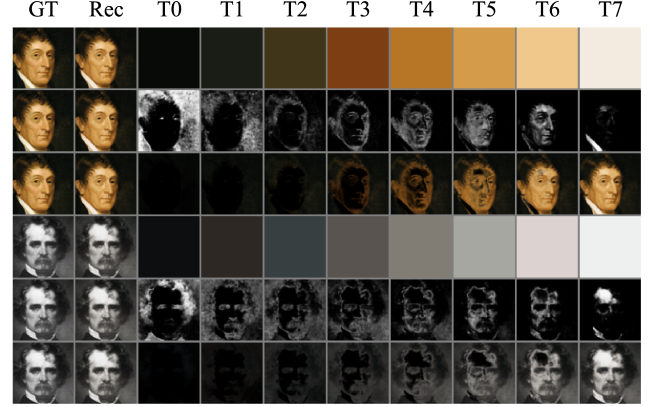
GroupPos. Inspired by group-based advantage normalization in GRPO (Yuan et al. 2025), we introduce GroupPos, a GAN loss that normalizes discriminator scores globally and applies soft non-negative gating. At each step, the generator produces a batch of B samples from different random z , and the discriminator assigns a score s_i to each. A global z-score is computed as $\text{adv}_i = (s_i - \mu)/\sigma$, where μ and σ are the mean and standard deviation across all B samples. The GroupPos loss is:

$$\mathcal{L}_G = -\frac{1}{B} \sum_{i=1}^B \frac{\text{softplus}(\text{adv}_i \cdot \tau)}{\tau} \cdot s_i, \quad (5)$$

where τ is a temperature parameter. The softplus gating produces weights in $(0, \infty)$ that are near-zero for samples far below the mean and approximately linear for samples above



(a) Reconstruction with ground truth above and reconstructions below.



(b) Per-token decomposition for two samples, showing token colors (rows 1 and 4), masks (rows 2 and 5), and cumulative build-up (rows 3 and 6).

Figure 2: MSE reconstruction and token decomposition on MetFaces.

the mean, so that below-mean samples contribute negligible positive weight rather than negative weight. Suppressing negative gradients is especially important when training from scratch, because a co-trained discriminator initially produces noisy scores, and a below-mean score often reflects reward model uncertainty rather than genuinely poor generation.

As training progresses, the discriminator adapts its decision boundary to the improving generator, providing progressively more informative feedback.

Experiments

Dataset and Setup

We conduct experiments on MetFaces (Karras et al. 2020) at 64×64 resolution. MetFaces (Karras et al. 2020) contains 1,336 face images from the Metropolitan Museum of Art, originally at 1024×1024 resolution, which we downsample to 64×64 .

We conduct three sets of experiments. MSE reconstruction on MetFaces verifies whether the softmax composition has sufficient capacity and whether tokens naturally specialize. GAN training with GroupPos evaluates generation quality. Token count extrapolation tests whether the model generalizes to different T at inference. All experiments run on a single NVIDIA RTX 3060 Laptop GPU with 6GB of memory.

MSE Reconstruction

We first verify TokenPainter’s reconstruction capability by training with MSE loss on fixed z -image pairs, where each



(a) Generated samples with $T = 8$, trained with GroupPos.



(b) Per-token decomposition for two samples from (a), showing token colors (rows 1 and 4), masks (rows 2 and 5), and cumulative build-up (rows 3 and 6).

Figure 3: GAN training results on MetFaces.

z is paired with one training image. Random resampling of z each epoch turns the task into unconditional generation, and the model collapses to the pixel-wise mean under MSE loss, so fixed pairs are necessary. Training on MetFaces for 500 epochs with bf16 mixed precision reaches a final MSE of 3.50×10^{-4} , PSNR of 35.55, and SSIM of 0.997.

Figure 2a shows reconstruction results with ground truth above and reconstructions below. The 8-token composition faithfully reproduces overall facial structure and broad color regions, but fine details such as facial blush, eye color, and hair tone variations are not fully recovered. This limitation stems from $T = 8$ providing only eight distinct colors and their corresponding masks. While three colors (RGB) suffice in principle, achieving this would require each token to carry global information spanning the entire canvas, which is prohibitively difficult to learn. Softmax composition can blend colors to produce intermediate tones, but since each token specializes in a local region, fine details that require

dedicated tokens can fall outside the coverage of eight masks.

Figure 2b presents the per-token decomposition for two samples. Both exhibit a similar decomposition pattern. In the first, T0 captures the background and hair, T1 supplements the background, T2 through T4 fill and refine the facial contour, T5 draws the main face color, T6 draws highlights across the entire face including the nose, cheekbones, and forehead, and T7 draws the white collar. In the second, T0 draws the hair and background, T1 through T4 supplement the facial contour, T5 and T6 add facial details, and T7 draws the collar and a forehead highlight. Without auxiliary losses, softmax competition alone drives tokens into complementary spatial roles, with later tokens refining details atop the broad structure established by earlier ones.

GAN Training

We train TokenPainter on MetFaces using the GAN framework described in Section 3.2, with the softplus discrimi-

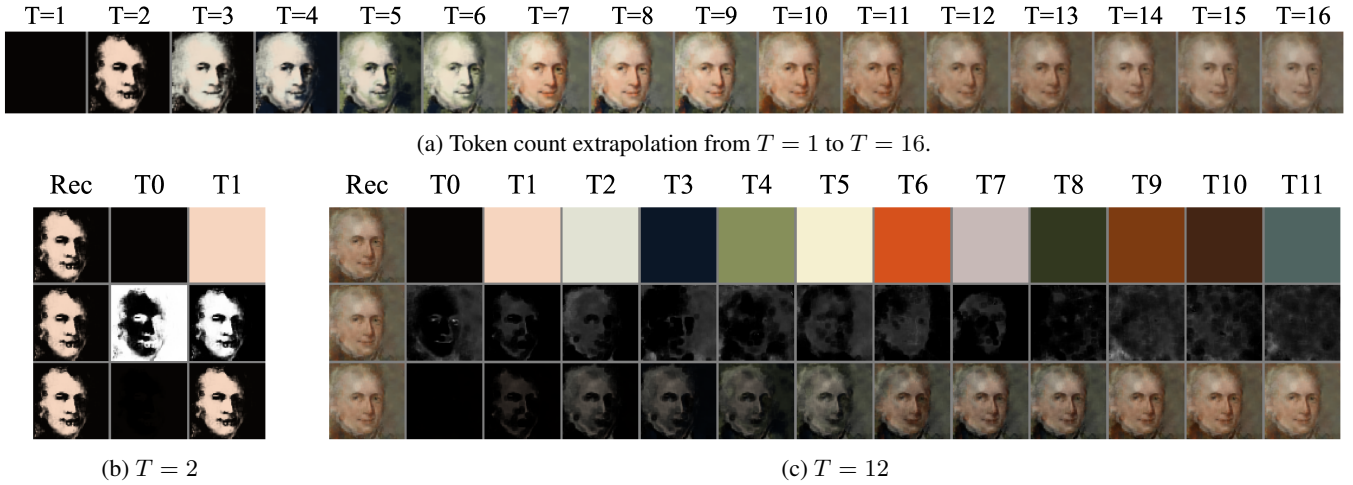


Figure 4: Token count extrapolation and decomposition.

nator loss, R1 and R2 gradient penalties ($\gamma=100$ every 4 steps), EMA with half-life 10 kimg, and GroupPos generator loss at temperature $\tau=2$. The generator and discriminator both use Adam with $\beta_1=0.0$, $\beta_2=0.9$ at constant learning rates of 4×10^{-4} and 2×10^{-4} , respectively. Training runs for 1000 kimg with bf16 mixed precision.

Figure 3a shows generated samples from the 00902k checkpoint with seed 42. While FID remains relatively high at 74.4, reflecting the limited dataset size, the generated faces are individually recognizable with coherent facial structure and diverse artistic styles, although fine details such as hair texture and facial contours are not perfectly rendered. Figure 3b presents the per-token decomposition for two selected samples. In the first sample, T0 draws the collar, facial contour, and eyes, T1 paints the overall face color, T2 draws the hair and face, T3 supplements the background and facial shadows, T4 fills partial background colors, T5 adds facial color, T6 draws the red lip color, and T7 completes the figure with final facial details. In the second sample, T0 draws the red collar, T1 renders facial shadows and eyes, T2 draws the background, T3 paints the face, T4 and T5 successively refine the facial contour, T6 supplements the background, and T7 draws the overall facial color. GAN-trained tokens exhibit a coarse-to-fine decomposition similar to MSE training, where early tokens establish structural regions and later tokens add fine chromatic details, driven solely by softmax competition without auxiliary decomposition losses.

Token Count Extrapolation

A key property of TokenPainter is that the model trained with a fixed T can be evaluated with a different number of tokens at inference time by adjusting RoPE positions, without any weight modification. We take the first sample from Figure 3b and evaluate the same GAN-trained $T=8$ checkpoint with token counts ranging from $T=1$ to $T=16$. Figure 4a shows the resulting images. At $T=1$, the model produces a single-color region with no discernible structure, because one token must represent the entire image and collapses to a uniform output. From $T=2$ to $T=8$, the face becomes progressively

sharper and more detailed as additional tokens contribute distinct colors and spatial masks, with $T=8$ achieving the best visual quality since it matches the training configuration. Beyond $T=8$, the overall facial shape is preserved but color saturation gradually dilutes and the image becomes increasingly blurred, consistent with RoPE positions beyond the training range receiving weaker attention weights and extra tokens contributing low-opacity overlapping masks. Notably, the subject identity and broad facial structure remain recognizable across the entire range from $T=2$ to $T=16$, demonstrating that the model’s representation generalizes gracefully to unseen token counts.

Figure 4b and Figure 4c show the per-token decomposition at $T=2$ and $T=12$, respectively. At $T=2$, the two tokens must partition the entire visual content, yet they produce a recognizable facial silhouette, confirming that even a minimal token budget captures global structure. At $T=12$, more colors are available than at $T=8$, which produces richer chromatic variation but also introduces visible blur as the additional four tokens contribute low-confidence masks that dilute the sharper masks of the core tokens. This U-shaped quality trend, with optimal clarity at the training T and degradation in both directions, mirrors the pattern observed under MSE training and suggests that RoPE-based length generalization is a robust property of the architecture rather than an artifact of the training objective.

We further investigate token-level modularity by selectively dropping tokens from the $T=8$ decomposition and re-normalizing softmax over the remaining subset. Figure 5 shows three such recompositions. Even-indexed tokens (Figure 5a) produce the best reconstruction among the three subsets, retaining clear facial features and background structure, which indicates that the even positions collectively cover the most informative spatial regions. Odd-indexed tokens (Figure 5b) yield the weakest output, with noticeably degraded facial contours, confirming that complementary visual roles are unevenly distributed across the sequence and that some tokens carry disproportionately more structural information. The combination of T1, T2, T5, and T6 (Figure 5c) achieves

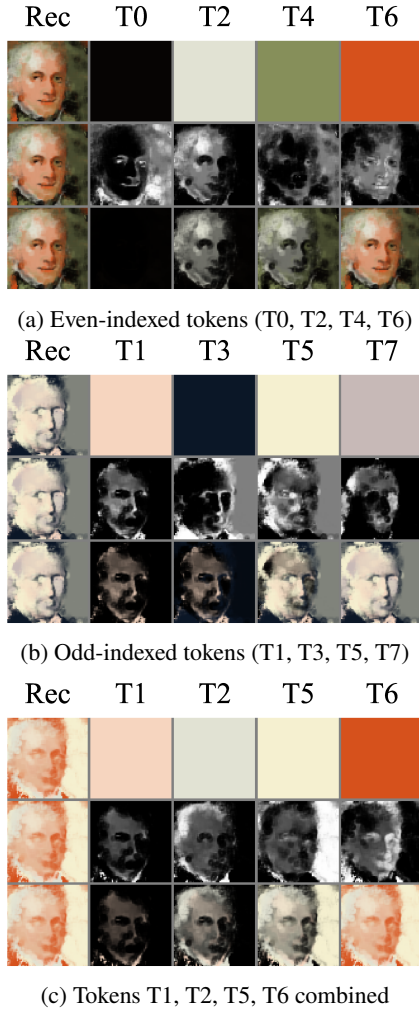


Figure 5: Token subset recomposition from $T = 8$, showing that sparse token subsets still produce recognizable outputs through softmax re-normalization.

intermediate quality, reconstructing a recognizable face despite using only half the tokens in an arbitrary subset. All three recompositions preserve the overall facial shape and subject identity, supporting the conclusion that each token independently encodes global spatial information through its full-canvas logits, and that softmax re-normalization dynamically redistributes spatial regions when tokens are removed. This emergent modularity is a direct consequence of per-token independent decoding combined with softmax competition, which forces every token to be a self-contained visual primitive capable of covering any region of the canvas.

Ablation Study

We compare two token budget configurations, $T = 8$ and $T = 6$, to quantify how the number of tokens affects generation quality under identical training conditions. Both models share the same architecture, hyperparameters, and GroupPos loss, differing only in the number of autoregressive steps. Figure 6 reports FID computed with 1000 generated sam-

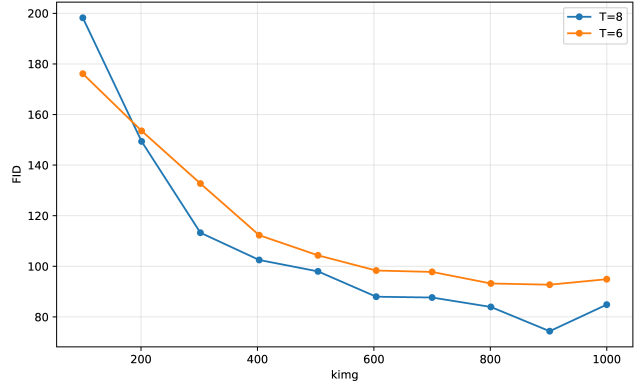


Figure 6: FID comparison between $T = 8$ and $T = 6$ on MetFaces.

ples against the full MetFaces training set at each 100k checkpoint. Both configurations exhibit steady and consistent FID improvement over the full 1000 kimg training run, with $T = 8$ decreasing from 198 at 100k to 74.4 at 902k, while $T = 6$ decreases from 176 to 92.7 over the same interval. The $T = 8$ model outperforms $T = 6$ at every checkpoint without exception, and the gap widens notably after 300k, confirming that additional tokens contribute meaningful representational capacity that translates directly into higher generation quality rather than adding redundant parameters. Both FID curves continue to descend at 902k without any sign of plateauing, which suggests that extending training beyond 1000 kimg or further increasing the token budget would yield additional improvements.

Discussion

TokenPainter’s emergent specialization arises from the combination of per-token independent decoding and softmax competition across tokens. Because each token is decoded without any cross-token communication, every token must independently encode its full visual contribution as a self-contained color and spatial mask, and the softmax over all token logits forces these independent proposals into a single convex combination that drives tokens toward non-overlapping and complementary spatial roles. The same mechanism that produces specialization during training also enables the subset recombination observed in Section 4.4, where softmax re-normalization over any selected subset of tokens dynamically redistributes spatial regions, allowing each remaining token to extend its coverage into vacated areas without any retraining or weight modification.

The GroupPos loss addresses a fundamental challenge of training generative models without reference images, where a co-trained discriminator initially produces noisy and poorly calibrated scores. By normalizing advantages globally via z-score and applying a softplus gate that produces strictly non-negative weights, GroupPos prevents the generator from being penalized for samples that score below the batch mean, which in early training often reflects discriminator uncertainty rather than genuinely poor generation quality. This design suppresses conflicting gradient signals that would

otherwise cause training collapse, allowing the generator to selectively reinforce the generations that the discriminator confidently favors. On the modest MetFaces dataset of only 1336 images, TokenPainter trained with GroupPos produces diverse and recognizable faces, demonstrating that the combination of per-token interpretable decoding and stable GAN training enables effective generation even under severe data constraints. The token count extrapolation and subset recomposition experiments further reveal that each token carries global spatial information, as evidenced by the model’s ability to generalize to unseen token counts and to reconstruct coherent outputs from arbitrary subsets of its trained token sequence.

Conclusion and Future Work

We presented TokenPainter, an autoregressive image generation model where each token independently decodes into an RGB color and spatial mask, making every generation step directly visible through softmax composition. Per-token independent decoding combined with softmax competition drives tokens to spontaneously specialize in complementary spatial regions without any auxiliary decomposition losses, producing an interpretable coarse-to-fine generation process. We introduced GroupPos, a GAN loss with global z-score advantage normalization and soft non-negative gating, which enables stable training from scratch on a single consumer GPU without reference images or large-scale pretraining. Experiments on MetFaces demonstrate that TokenPainter generates diverse and recognizable faces, generalizes to different token counts at inference via RoPE-based length extrapolation, and supports token-level modularity where arbitrary subsets of tokens recombine into structurally consistent outputs. These properties suggest that treating tokens as visible drawing primitives is a promising direction for interpretable image generation. In future work, we plan to scale TokenPainter to higher resolutions and natural image datasets by increasing the token budget, which our extrapolation results suggest is feasible, and to explore conditional generation where dedicated groups of tokens model different semantic attributes.

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